

International Economic Relations

ECON 308

External Economies of Scale and the International Location of
Production

(Chapter 7)

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Introduction (1 of 4)

- The models of comparative advantage thus far assumed **constant returns to scale**:
 - When inputs to an industry increase at a certain rate, output increases at the same rate.
 - If inputs to an industry were doubled, industry output would double as well.

Introduction (2 of 4)

- But there may be **increasing returns to scale** or **economies of scale**:
 - This means that when inputs to an industry increase at a certain rate, output increases at a faster rate.
 - A larger scale is more efficient: the cost per unit of output falls as a firm or an industry increases output.

Introduction (3 of 4)

- For example, suppose an industry produces widgets using only one input, labor.
- Consider how the amount of labor required depends on the number of widgets produced.
- The presence of economies of scale may be seen from the fact that
 - doubling the input of labor more than doubles the industry's output.
 - the average amount of labor used to produce each widget is less when the industry produces more.

Table 7.1 Relationship of Input to Output for a Hypothetical Industry

Output	Total Labor Input	Average Labor Input
5	10	2
10	15	1.5
15	20	1.333333
20	25	1.25
25	30	1.2
30	35	1.166667

Introduction (4 of 4)

- Mutually beneficial trade can arise as a result of economies of scale.
- International trade permits each country to produce a limited range of goods without sacrificing variety in consumption.
- With trade, a country can take advantage of economies of scale to produce more efficiently than if it tried to produce everything for itself.

Economies of Scale and Market Structure (1 of 2)

- Economies of scale could mean either that larger firms or a larger industry would be more efficient.
- **External economies of scale** occur when the cost per unit of output depends on the **size of the industry**.
- **Internal economies of scale** occur when the cost per unit of output depends on the **size of the firm**.

Economies of Scale and Market Structure (2 of 2)

- Both external and internal economies of scale are important causes of international trade.
- They have different implications for the structure of industries:
 - An industry where economies of scale are purely external will typically consist of many small firms and be perfectly competitive.
 - Internal economies of scale result when large firms have a cost advantage over small firms, causing the industry to become imperfectly competitive.

The Theory of External Economies (1 of 6)

- This chapter deals with a model of external economies; the next chapter will cover internal economies.
- Many modern examples of industries that seem to be powerful external economies:
 - In the United States, the semiconductor industry is concentrated in Silicon Valley, investment banking in New York, and the entertainment industry in Hollywood.

The Theory of External Economies (2 of 6)

- In developing countries such as China, external economies are pervasive in manufacturing.
 - One town in China produces most of the world's underwear production, another produces nearly all cigarette lighters.
- External economies played a key role in India's emergence as a major exporter of information services.
 - Indian information services companies are still clustered in Bangalore.

The Theory of External Economies (3 of 6)

- For a variety of reasons, concentrating production of an industry in one or a few locations can reduce the industry's costs, even if the individual firms in the industry remain small.
- External economies may exist for a few reasons:

The Theory of External Economies (4 of 6)

1. **Specialized equipment or services** may be needed for the industry, but are only supplied by other firms if the industry is large and concentrated.
 - For example, Silicon Valley in California has a large concentration of silicon chip companies, which are serviced by companies that make special machines for manufacturing silicon chips.
 - These machines are cheaper and more easily available there than elsewhere.

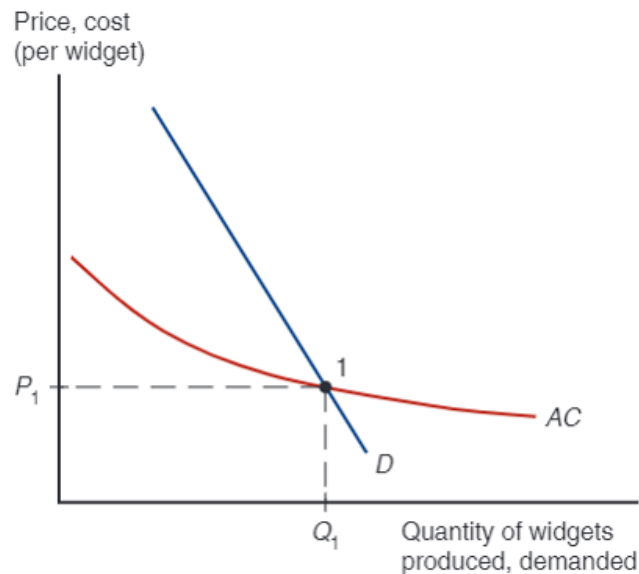
The Theory of External Economies (5 of 6)

2. **Labor pooling:** a large and concentrated industry may attract a pool of workers, reducing employee search and hiring costs for each firm.
3. **Knowledge spillovers:** workers from different firms may more easily share ideas that benefit each firm when a large and concentrated industry exists.

The Theory of External Economies (6 of 6)

- Represent external economies simply by assuming that the larger the industry, the lower the industry's costs.
- There is a **forward-falling supply curve**: the larger the industry's output, the lower the price at which firms are willing to sell.
- Without international trade, the unusual slope of the supply curve doesn't matter much.

Figure 7.1 External Economies and Market Equilibrium

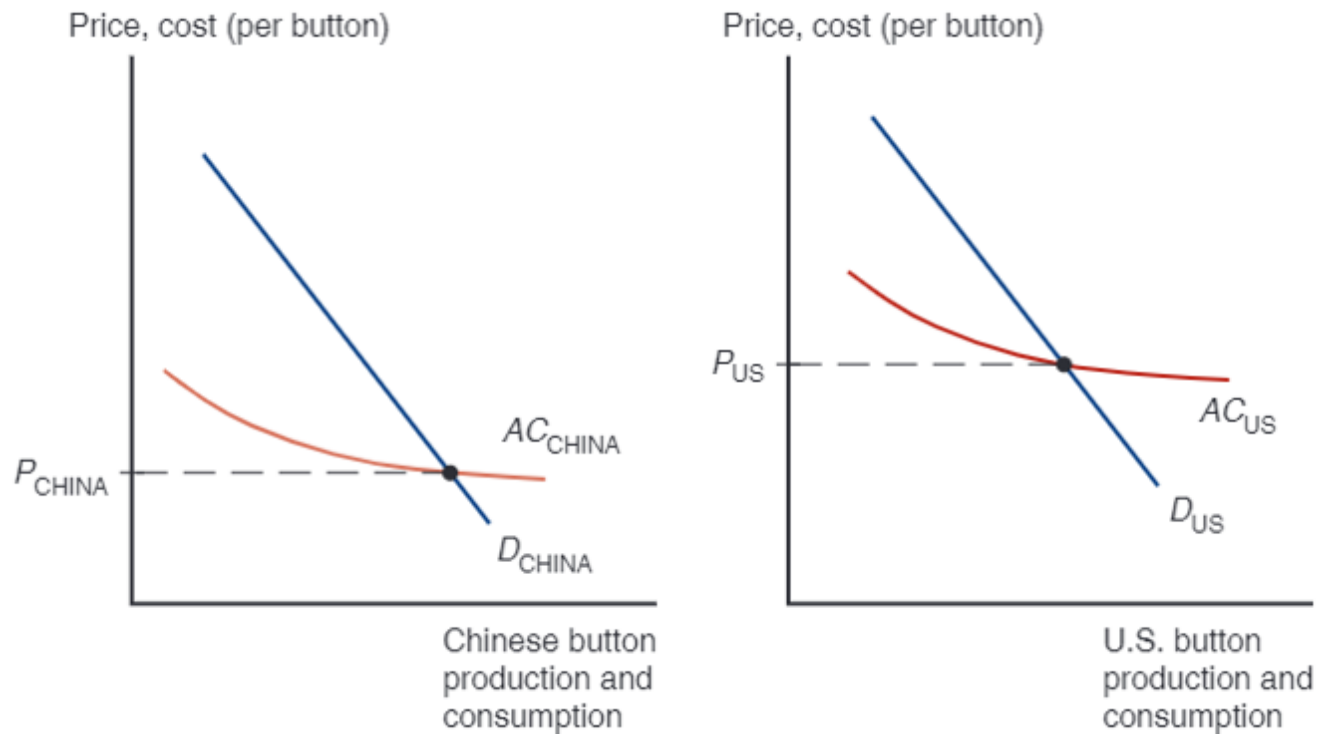


When there are external economies of scale, the average cost of producing a good falls as the quantity produced rises. Given competition among many producers, the downward-sloping average cost curve AC can be interpreted as a **forward-falling supply curve**. As in ordinary supply-and-demand analysis, market equilibrium is at point 1, where the supply curve intersects the demand curve, D . The equilibrium level of output is Q_1 , the equilibrium price P_1 .

External Economies and International Trade (1 of 10)

- Prior to international trade, equilibrium prices and output for each country would be at the point where the domestic supply curve intersects the domestic demand curve.
- Suppose Chinese button prices in the absence of trade would be lower than U.S. button prices.

Figure 7.2 External Economies Before Trade



In the absence of trade, the price of buttons in China, P_{CHINA} , is lower than the price of buttons in the United States, P_{US} .

External Economies and International Trade (2 of 10)

- What will happen when the countries open up the potential for trade in buttons?
- The Chinese button industry will expand, while the U.S. button industry will contract.
- This process feeds on itself: As the Chinese industry's output rises, its costs will fall further; as the U.S. industry's output falls, its costs will rise.
- In the end, all button production will be in China.

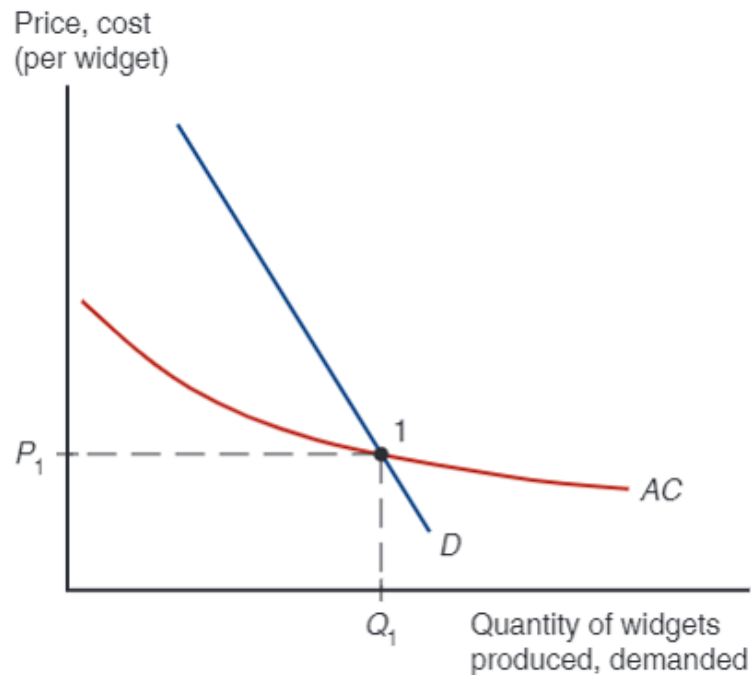
External Economies and International Trade (3 of 10)

- How does this concentration of production affect prices?
- Chinese button prices were lower than U.S. button prices before trade.
- Because China's supply curve is forward-falling, increased production as a result of trade leads to a button price that is lower than the price before trade.
- **Trade leads to prices that are lower than the prices in either country before trade!**

External Economies and International Trade (4 of 10)

- This is very different from the implications of models without increasing returns.
- In the standard trade model, relative prices converge as a result of trade.
- If cloth is relatively cheap in the home country and relatively expensive in the foreign country before trade opens, the effect of trade was to raise cloth prices in Home and reduce them in Foreign.
- **With external economies, by contrast, the effect of trade is to reduce prices everywhere.**

Figure 7.3 Trade and Prices



When trade is opened, China ends up producing buttons for the world market, which consists both of its own domestic market and of the U.S. market. Output rises from Q_1 to Q_2 , leading to a fall in the price of buttons from P_1 to P_2 , which is lower than the price of buttons in either country before trade.

External Economies and International Trade

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- What might cause one country to have an initial advantage from having a lower price?
- One possibility is comparative advantage due to underlying differences in technology and resources.
- If external economies exist, however, the pattern of trade could be due to historical accidents:
 - Countries that start as large producers in certain industries tend to remain large producers even if another country could potentially produce more cheaply.

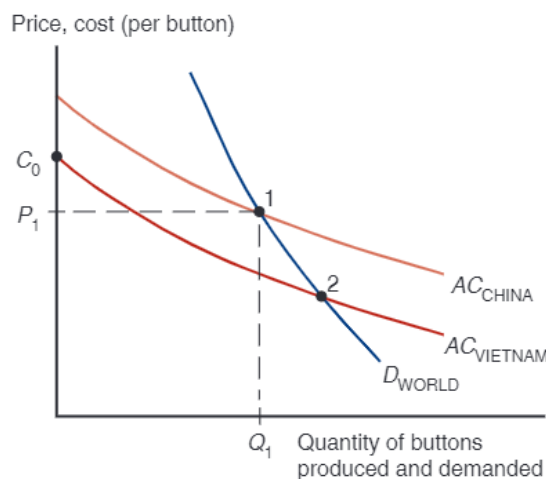
External Economies and International Trade (6 of 10)

- A tufted blanket, crafted as a wedding gift by a 19th-century teenager, gave rise to the cluster of carpet manufacturers around Dalton, Georgia.
- Silicon Valley may owe its existence to two Stanford graduates named Hewlett and Packard who started a business in a garage there.

External Economies and International Trade (7 of 10)

- Assume that the Vietnamese cost curve lies below the Chinese curve because Vietnamese wages are lower than Chinese wages.
- At any given level of production, Vietnam could manufacture buttons more cheaply than China.
- One might hope that this would always imply that Vietnam will in fact supply the world market.
- But this need not always be the case if China has enough of a head start.
- **No guarantee that the right country will produce a good that is subject to external economies.**

Figure 7.4 The Importance of Established Advantage



The average cost curve for Vietnam, AC_{VIETNAM} , lies below the average cost curve for China, AC_{CHINA} . Thus Vietnam could potentially supply the world market more cheaply than China. If the Chinese industry gets established first, however, it may be able to sell buttons at the price P_1 , which is below the cost C_0 that an individual Vietnamese firm would face if it began production on its own. So a pattern of specialization established by historical accident may persist even when new producers could potentially have lower costs.

External Economies and International Trade (8 of 10)

- Trade based on external economies has an ambiguous effect on national welfare.
 - There will be gains to the world economy by concentrating production of industries with external economies.
 - **It's possible that a country is worse off with trade than it would have been without trade:** a country may be better off if it produces everything for its domestic market rather than pay for imports.

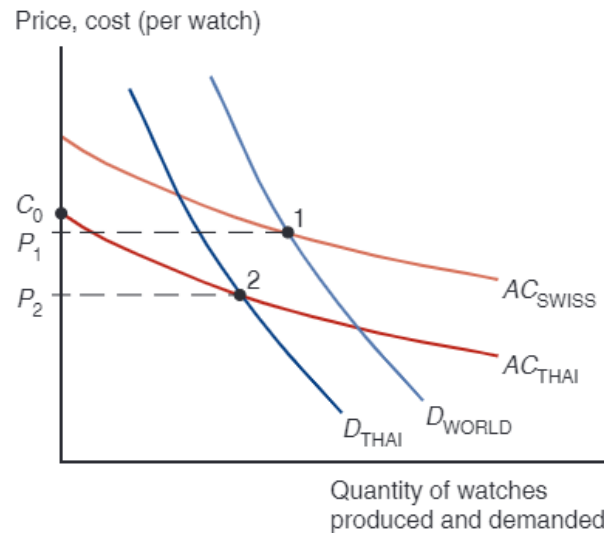
External Economies and International Trade (9 of 10)

- Imagine that Thailand could make watches more cheaply, but Switzerland got there first.
- The price of watches could be lower in Thailand with no trade.
- Trade could make Thailand worse off, creating an incentive to protect its potential watch industry from foreign competition.
- What if Thailand reverts to autarky?

External Economies and International Trade (10 of 10)

- Note that it's still to the benefit of the **world** economy to take advantage of the gains from concentrating industries.
- Each country wanting to reap the benefits of housing an industry with economies of scale creates trade conflicts.
- Overall, it's better for the world that each industry with external economies be concentrated **somewhere**.

Figure 7.5 External Economies and Losses from Trade



When there are external economies, trade can potentially leave a country worse off than it would be in the absence of trade. In this example, Thailand imports watches from

Switzerland, which is able to supply the world market (D_{WORLD}) at a price (P_1) low enough to block entry by Thai producers, who must initially produce the watches at cost C_0 . Yet if Thailand were to block all trade in watches, it would be able to supply its domestic market (D_{THAI}) at the lower price, P_2 .